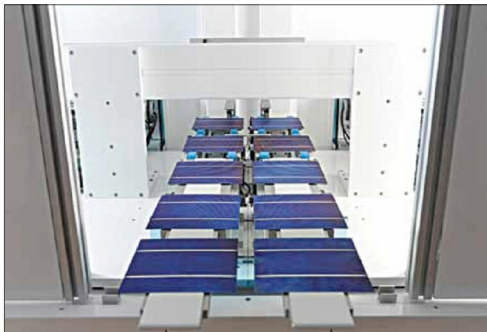


PV Cell Manufacturers Look Beyond Cost of Ownership

Evaluating the total cost of ownership is no longer sufficient to assess the quality of a manufacturing tool, which is a great investment. Such an investment should be evaluated in the light of its potential for generating differential value along the years



COURTESY: APPLIED MATERIALS

Solar is the next great energy frontier. Interest in renewable energy sources has never been higher, and manufacturing process technology is advancing at a rapid pace to meet the demands of the photovoltaic market. Sunlight is free, but the technology to convert it into usable energy can be very expensive. Driving the cost per watt of solar energy down will be key to achieving grid parity in major markets in order to expand solar PV application worldwide—and to profit.

In the beginning of the solar-energy era, the cost of manufacturing a solar cell was tied directly to the capital cost of manufacturing equipment. Price was a key element of a

fairly standard process where there was not much differentiation among original equipment manufacturers and the game was mostly an equipment price battle. Companies did the math to convert equipment cost into productivity, and quickly informed suppliers that the bottom line was more than the market would bear. New technologies are often very expensive—the cost of the first personal computers was as much as a mortgage payment—limiting the number of potential end customers. Just as innovation in chip manufacturing has radically reduced the size and cost of PCs to the point where most people can afford a computer or smartphone, the solar market was ripe for innovation that would reduce costs and lead the way to grid parity.

Total cost of ownership

A new metric has emerged that is a more comprehensive way than simple capital expenditure to compare and evaluate the cost of different solutions. Total cost of ownership (TCO), sometimes shortened to cost of ownership, was developed by The Gartner Group in the 1980s. It rapidly became a standard methodology providing valuable insight into the direct and indirect costs of a system over a predetermined period of time. For high-tech companies, this period is generally five years, because that is assumed to be the amount of time that a piece of equipment will remain relevant before a new technology is needed.

In addition to the capital procurement cost of tools, additional factors should be considered in calculating TCO. There are dozens of these, and a true side-by-side comparison can be very difficult to calculate. Semiconductor companies can seek guidance from SEMI—the global industry association for certain semiconductor manufacturing supply chains. Their document SEMI E35-0307, Guide to Calculate Cost of Ownership (COO) Metrics for Semiconductor Manufacturing Equipment, recommends quantifying many factors within three main categories:

1. Fixed costs (essentially depreciation for property, plant and equipment)
2. Recurring costs (mostly direct materials, consumables, maintenance and labour)
3. Yield costs (essentially scrap and mean performance)